

Homework 6

Ravi Kini

November 28, 2025

Exercise 1

Since $\phi(17) = 16$, we have that $\text{ord}_{17} 9 \mid 16$. Considering the powers of 9 that divide 16:

k	$9^k \bmod 17$
1	9
2	13
4	16
8	16

Consequently, $\text{ord}_{17} 9 = 8$.

Exercise 2

Since $\phi(18) = 6$, for r to be a primitive root modulo 18, $\text{ord}_{18} r = 6$. The reduced residues are 1, 5, 7, 11, 13, 17. Considering their powers that divide 6:

k	$1^k \bmod 18$	$5^k \bmod 18$	$7^k \bmod 18$	$11^k \bmod 18$	$13^k \bmod 18$	$17^k \bmod 18$
1	1	5	7	11	13	17
2		7	13	13	7	1
3		17	1	17	1	
6		1		1		

The primitive roots modulo 18 are then $\{5, 11\}$.

Exercise 3

Part (a)

Let $m \in \mathbb{N}$ and a, b such that $(a, m) = (b, m) = 1$ and $(\text{ord}_m a, \text{ord}_m b) = 1$. Let $\text{ord}_m a = i$, $\text{ord}_m b = j$, and $\text{ord}_m ab = k$. Then:

$$\begin{aligned}(ab)^{ij} &= a^{ij} \cdot b^{ij} \\ &= (a^i)^j \cdot (b^j)^i \\ &= 1 \cdot 1 = 1\end{aligned}$$

Consequently, $k | ij$. Then:

$$\begin{aligned}(ab)^k &= 1 \pmod{m} \\ (ab)^{kj} &= 1 \pmod{m} \\ a^{kj} \cdot b^{kj} &= \\ a^{kj} \cdot (b^j)^k &= \\ a^{kj} \cdot 1^k &= \\ a^{kj} &= 1\end{aligned}$$

Consequently, $i | kj$. Since $(i, j) = 1$, this means that $i | k$. We can similarly show that $j | k$, and so $ij | k$. As a result, $ij = k$, and so $\text{ord}_m ab = \text{ord}_m a \cdot \text{ord}_m b$.

Part (b)

Consider the case where $m = 7$, $a = 2$, and $b = 4$. By considering the powers of a and b , we see that $\text{ord}_7 2 = 6$ and $\text{ord}_7 4 = 3$. Consequently, $(\text{ord}_m a, \text{ord}_m b) \neq 1$. Further, $\text{ord}_7 8 = 1$, and so $\text{ord}_m ab \neq \text{ord}_m a \cdot \text{ord}_m b$.

Exercise 4

Let p be an odd prime. Suppose r is a primitive root modulo p . Then $\text{ord}_p r = \phi(p) = p - 1$. Suppose q is a prime divisor of $p - 1$ and $r^{(p-1)/q} = 1 \pmod{p}$. Note that $(p-1)/q < p$. Then $\text{ord}_p r = (p-1)/q$ and so r cannot be a primitive root. This is a contradiction; therefore $r^{(p-1)/q} \neq 1 \pmod{p}$ for all prime divisors q of $p - 1$. Conversely, suppose $r^{(p-1)/q} \neq 1 \pmod{p}$ for all prime divisors q of $p - 1$. Let $d = \text{ord}_p r$. By Fermat's Little Theorem, $d \mid (p - 1)$. Suppose $d < p - 1$, and let q' be some prime factor of $(p - 1)/d$, which must also be a factor of $p - 1$. Note that $d \mid (p - 1)/q'$. However, letting $e = (p - 1)/q'd$, this means that $r^{(p-1)/q'} = (r^d)^e = 1^e = 1 \pmod{p}$. This is a contradiction; therefore $\text{ord}_p r = p - 1$ and so r is a primitive root.

Exercise 5

Let $m \in \mathbb{N}$, and let r be a primitive root modulo m . Let $\bar{r} = r^{-1} \pmod{m}$, and let $d = \text{ord}_m \bar{r}$. Since $\text{ord}_m r = \phi(m)$:

$$\begin{aligned}(r\bar{r})^{\phi(m)} &= 1^{\phi(m)} \pmod{m} \\ r^{\phi(m)} \cdot \bar{r}^{\phi(m)} &= 1 \pmod{m} \\ \bar{r}^{\phi(m)} &= \end{aligned}$$

Consequently, $d \mid \phi(m)$. Then:

$$\begin{aligned}\bar{r}^d &= 1 \pmod{m} \\ r^d \cdot \bar{r}^d &= r^d \pmod{m} \\ (r\bar{r})^d &= \\ 1^d &= 1 = \end{aligned}$$

Consequently, $\phi(m) \mid d$, and so $\text{ord}_m \bar{r} = d = \phi(m)$, which means that $\bar{r}$ is a primitive root modulo m .